

8.4.3 Increase the Number of Replicates

In some cases, the obtained MU is higher than expected and makes it difficult to decide with a sufficient level of confidence. A classic example is the compliance of a sample containing traces when the specification limit is close to the limit of quantification. A possible way to overcome this shortcoming is to work with replicates and make the decision based on an average result. This approach impacts the MU and implies some corrections for this way of expressing the result.

First of all, the term *replicate* needs to be defined. At what stage of the analytical operating procedure does the replication begin? When are different test portions sampled, when the extract (or the reconstituted sample) is injected into the measuring instrument, or at another moment? The replicates done at different starting points do not have the same uncertainty, since diverse sources of uncertainty may be triggered. The diagram presented in Figure 8.14 is an attempt to illustrate the multiplicity of replicate definitions and formalize this concept of replicate.

It is now accepted to divide an analytical process into three complementary parts: pre-analytical, analytical, and post-analytical. Replication itself is limited to the analytical part. The US-Pharmacopeia (USP) document [17] proposes also

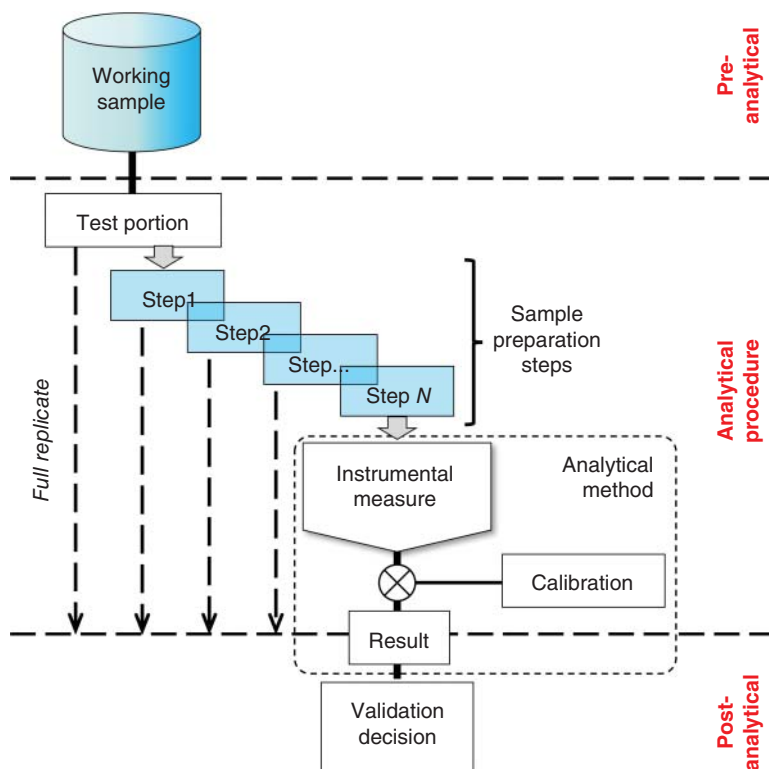


Figure 8.14 Four possible replicate definitions according to the sample preparation starting step among the sample preparation steps.

to make a division, within the analytical part, between the analytical procedure, which includes the sample preparation steps, and the analytical method, which corresponds to the actual instrumental measurement, the calibration, and the result calculation.

This partition is helpful to better define replication strategies intended to reduce MU in drug monitoring [18]. In Figure 8.14 the vertical dashed arrows illustrate four distinct types of replicates that can be performed starting at various stages of a sample preparation. The replication procedure, labeled “*Full replicate*” on the figure, starts from a new test portion and consists of applying the complete analytical operating procedure. This is the most interesting because it represents the highest level of independence that can be obtained between two replicates done on the same working sample.

Unfortunately, the word “sample” is often indiscriminately and carelessly used to refer either to the entity that is received (or collected) by the laboratory for analysis or to the resulting preparation, such as an extract or reconstituted material after sample preparation, which is introduced into the measuring instrument. In the previous Section, 8.3, about sampling uncertainty, some definitions were proposed to avoid this confusion. It is essential that all replicates be obtained on the same sample, i.e. the same entity received by the laboratory.

Let us note Z_n with $1 \leq n \leq N$ the replicates used to calculate an average result. Two situations must be clearly distinguished while in each case, the result MU is differently estimated when dealing with the average of the replicates:

- Measures under repeatability condition. Replicates are performed on the same day by the same operator in the same series. It can be assumed they have the same standard uncertainty, denoted $u(Z_n)$.
- Measures under intermediate precision conditions. A common way to make replicates when a sample is declared nonconforming to check the consistency of the decision is to repeat one measurement (or several measurements) later during another sequence or series.

8.4.4 Replication under Repeatability Condition

A reliable replicate should be performed from a new test portion, as described in Figure 8.14 to include as many sources of uncertainty as possible under repeatability conditions. It is also the simplest situation to compute the result MU. If the uncertainty function of the method has been established as described in Section 7.5, it is possible to estimate $u(Z_n)$ for any Z_n . According to the GUM, two measurement models should be considered for an average of replicates. They are denoted here $\overline{Z}_A$ and $\overline{Z}_B$.

Model A

$$\overline{Z}_A = \frac{\sum_{n=1}^N Z_n}{N} \quad (8.18)$$

Model B

$$\overline{Z}_B = \frac{\sum_{n=1}^N Z_n}{N} + E_r \quad (8.19)$$

Model B includes the idea that measurements, although obtained under repeatability conditions, are not all equal, and a residual random error accounts for the differences. The random variable E_r reflects these random variations between replicates. When these models are developed according to the law of uncertainty propagation, the following MU estimate is obtained for each model.

Standard variance of model A

$$u^2(\overline{Z}_A) = \frac{\sum_{n=1}^N u^2(Z_n)}{N} \quad (8.20)$$

Standard uncertainty of model A

$$u(\overline{Z}_A) = \sqrt{u^2(\overline{Z}_A)} \quad (8.21)$$

Standard variance of E_r

$$u^2(E_r) = \frac{\sum_{n=1}^N (Z_n - \overline{Z}_A)^2}{\frac{N-1}{N}} \quad (8.22)$$

Standard uncertainty of model B

$$u(\overline{Z}_B) = \sqrt{u^2(\overline{Z}_A) + u^2(E_r)} \quad (8.23)$$

A classic statistical property is used here, namely: if a random variable is the sum of N random variables of the same standard deviation σ , its standard deviation is $\frac{\sigma}{\sqrt{N}}$. This property allows us to set the following table, which shows the possible reduction gain on the MU when the number of replicates increases from one single measure. For example, to reduce MU by one-half, it is necessary to do at least four replicates.

N	$1/\sqrt{N}$	Gain (%)
2	0.71	29
3	0.58	42
4	0.50	50
5	0.45	55
8	0.35	65
10	0.32	68

The results of the THEOPHYLLINE study will serve as an example. The values obtained for eight replicates made on the same sample are reported in Table 8.8. The column “Mean” corresponds to the provisional calculation of the average; for example, 0.1515 corresponds to 2 replicates. The standard uncertainty function of the method is provided in Section 7.5.2 and illustrated in Figure 7.8a. It is a power

Table 8.8 THEOPHYLLINE – influence of the number of replicates.

Measure number	Value	Mean	Model A		Model B	
			$u^2(\overline{Z_A})$	$u(\overline{Z_A}) \times 10^3$	$u^2(e_r)$	$u(\overline{Z_B}) \times 10^3$
1	0.154			21.150		21.150
2	0.149	0.1515	$2.17 \cdot 10^{-4}$	14.728	$6.25 \cdot 10^{-6}$	14.939
3	0.151	0.1513	$1.45 \cdot 10^{-4}$	12.025	$2.11 \cdot 10^{-6}$	12.113
4	0.155	0.1523	$1.08 \cdot 10^{-4}$	10.414	$1.90 \cdot 10^{-6}$	10.505
5	0.150	0.1518	$8.68 \cdot 10^{-5}$	9.315	$1.34 \cdot 10^{-6}$	9.387
6	0.148	0.1512	$7.23 \cdot 10^{-5}$	8.503	$1.29 \cdot 10^{-6}$	8.579
7	0.152	0.1513	$6.20 \cdot 10^{-5}$	7.873	$9.39 \cdot 10^{-7}$	7.932
8	0.151	0.1513	$5.42 \cdot 10^{-5}$	7.364	$7.05 \cdot 10^{-7}$	7.412

function with the following coefficients: constant $a = 0.0907$ and power $b = 0.778$. Thus, the standard uncertainty of the first measure $0.154 \mu\text{g/l}$ is $u(Z) = 21.150 \cdot 10^{-3} \mu\text{g/l}$.

All MU values are multiplied by 10^3 to be more readable. As expected, Table 8.8 shows that the MU of the average result regularly decreases from $21 \cdot 10^{-3}$ to 7.10^{-3} , when the number of replicates increases from 1 to 8, i.e. a final 67% reduction. Obviously, the cost of this procedure can be high, and a cost/benefit study may help to decide the adequate replicate number. When comparing the estimates obtained with model A or B, the correction introduced in model B only slightly modifies the MU estimate.

Further refinement can be introduced in the measurement model of a result corresponding to an average of replicates because replicates are made under repeatability conditions and consequently correlated. The law of propagation of uncertainty can consider this kind of correlation as illustrated by Equation (6.18) before it is simplified in Equation (6.21). The difficulty is to evaluate practically this correlation.

8.4.5 Replication under Intermediate Precision Condition

As previously explained, when a measurement value is not as satisfactory as expected (e.g. when a whole production would not be marketable because of one nonconforming result), it is quite common to do a replicate to infirm, confirm, strengthen, or weaken the conclusion. Sometimes, this operation is repeated several times to achieve a certain degree of confidence for the decision. If the new analysis does not belong to the same series, the between-series effect must be added, which implies that an estimate is available.

To explain the calculation, the usual notation is utilized. Let us assume that n replicates are done; the uncertainty function is available, and an estimate $u(Z_n)$ is available for each. This results in the following formulas.

Mean (result)

$$\bar{Z} = \frac{\sum_{n=1}^N Z_n}{I} \quad (8.24)$$

Within-replicate variance

$$s_r^2 = \frac{\sum_{n=1}^N u^2(Z_n)}{N} \quad (8.25)$$

Between-replicates variance

$$s_B^2 = ? \quad (8.26)$$

Standard variance of the mean

$$u^2(\bar{Z}) = \frac{s_B^2 + s_r^2}{N} \quad (8.27)$$

This is an uncomfortable situation when no estimate of the between-replicates variance is available, but it is common. Therefore, it is a risky practice to do replicates under intermediate precision and compute an average to make a final decision. Decision rules come with a sampling plan. If only one measurement is required to decide whether a lot is conforming, this rule must be respected. It is always possible to perform confirmatory analyses, but it is not correct to use the average for the decision unless special regulations exist.

References

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